

Lecture 4:

Multiparameter Models

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Announcements

- Today: Chapter 3
- Next: Chapter 4 (can skim) and Chapter 5

(Hierarchical Modeling)

• Talk due Sunday midnight.

Bayesian inference in the normal model

- Assume $y_1, \dots, y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant ^{e.g.}
- Let's start with a Jeffreys' (improper prior): $p(\mu) \propto \text{const}$
- What is the posterior distribution $p(\mu \mid y_1, \dots, y_n, \sigma^2)$?

Jeff. $\sqrt{I(\mu)} = \sqrt{\frac{n}{\sigma^2}} \propto \text{const.}$ (improper!)

$$P(\mu \mid y_1, \dots, y_n) \propto L(\mu) P(\mu) \propto L(\mu)$$

$$\propto \prod_{i=1}^n e^{-\frac{(y_i - \mu)^2}{2\sigma^2}} \propto e^{-\sum \frac{(y_i - \mu)^2}{2\sigma^2}}$$

$$\propto e^{-\frac{-n\mu^2 + 2(\sum y_i)\mu - \sum y_i^2}{2\sigma^2}} \quad \begin{matrix} 1/n \\ 1/n \end{matrix}$$

$$\propto e^{-\frac{\mu^2 + 2\bar{y}\mu}{2\sigma^2/n}} \times e^{\cancel{\frac{\sum y_i^2}{n}}}$$

$$\propto e^{-\frac{(\mu - \bar{y})^2}{2\sigma^2/n}} \times \text{const} \quad E[\mu | y_1, \dots, y_n]$$

Bayes $\mu | y_1, \dots, y_n, \sigma^2 \sim N(\bar{y}, \frac{\sigma^2}{n})$

$\text{Var}(\mu | y_1, \dots, y_n)$

Freq

$$\hat{\mu}_{MLE} = \bar{y} \sim N(\mu, \frac{\sigma^2}{n})$$

$$\sqrt{\sigma^2/n} (\mu - \bar{y}) \sim N(0, 1)$$

$$E[\mu | y_1, \dots, y_n] = \bar{y} = \hat{\mu}_{MLE}$$

Confidence Interval equiv. Bayes Credible
Interpretable.

Bayesian inference in the normal model

- Assume $y_1, \dots, y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant
- The normal prior distribution is conjugate for μ in the normal sampling model
- Sampling distribution, prior distribution and posterior distribution are all normal.
- Assume the prior is $p(\mu) \sim N(\mu_0, \sigma^2 / \kappa_0)$
- What are the parameters of the posterior $p(\mu \mid y_1, \dots, y_n, \sigma^2)$?

Prior Parameter

Normal-Normal model.

γ^2

$$P(\mu | y_1, \dots, y_n, \sigma^2) \propto L(\mu) P(\mu)$$

$$\propto \underbrace{e^{-\frac{(\mu - \bar{y})^2}{2\sigma^2/n}}}_{L(\mu)} \underbrace{e^{-\frac{(\mu - \mu_0)^2}{2\sigma^2/k_0}}}_{P(\mu)}$$

$$\propto \exp \left[-\frac{1}{2} \left(\frac{n+k_0}{\sigma^2} \right) \mu^2 - 2 \left(\frac{n\bar{y}}{\sigma^2} + \frac{k_0 \mu_0}{\sigma^2} \right) \mu + \text{consts} \right]$$

$$\propto \exp \left[-\frac{1}{2} \left(\frac{n+k_0}{\sigma^2} \right) \left(\mu - \frac{\frac{n\bar{y}}{\sigma^2} + \frac{k_0 \mu_0}{\sigma^2}}{\frac{n+k_0}{\sigma^2}} \right)^2 \right]$$

$$\mu | y_1, \dots, y_n \sim N \left(w\bar{y} + (1-w)\mu_0, \frac{\sigma^2}{n+k_0} \right)$$

$$E[\mu | y_1, \dots, y_n] = \underbrace{w\bar{y}}_{MLE} + (1-w) \underbrace{\mu_0}_{\substack{\uparrow \\ \text{prior} \\ \text{mean}}}$$

$$w = \frac{\frac{n}{\sigma^2}}{\frac{n+k_0}{\sigma^2}} = \frac{n}{n+k_0}$$

"pseudo" sample size

"Total" = pseudo + Actual.
Sample size.

$$\text{Var}(\mu | y_1, \dots, y_n) = \frac{\sigma^2}{n+k_0}$$

A conjugate prior for the normal likelihood

- The normal distribution is conjugate for the normal likelihood
 - Often called the “normal-normal model”
- $Y_i \sim N(\mu, \sigma^2)$ and $\mu \sim N(\mu_0, \sigma^2 / \kappa_0)$ implies that the posterior distribution $p(\mu \mid y)$ is also normally distributed:

$$\mu \mid Y \sim N(\mu_n, \tau_n^2)$$

$E[\mu \mid y_1, \dots, y_n]$

$\text{Var}(\mu \mid y_1, \dots, y_n)$

$$\text{where } \mu_n = \frac{\frac{\kappa_0}{\sigma^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}} \text{ and } \tau_n^2 = \frac{1}{\frac{\kappa_0}{\sigma^2} + \frac{n}{\sigma^2}}$$

The posterior mean and pseudo-counts

$$\mu_n = \frac{\frac{1}{\tau^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}} \mu_0 + \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}} \bar{y}$$

Fisher weight = $(1 - w)\mu_0 + w\bar{y}$

Fisher Information.

where $w = \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$

Prior variance: $\tau^2 = \frac{\sigma^2}{K_0}$

Prior precision

Precision-weighted.

$$y_1, \dots, y_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

$\nwarrow$ known

Conjugate Prior: $\mu \sim N(\mu_0, \frac{\sigma^2}{K_0})$

$$\Rightarrow P(\mu | y_1, \dots, y_n, \sigma^2) \sim N(w\bar{y} + (1-w)\mu_0, \frac{\sigma^2}{K_0 + n})$$

$$w = \frac{n}{n + K_0}$$

Known mean, unknown variance

- Assume we have n mean-zero normal observations with variance σ^2

- Define $d_i = (y_i - \mu)$ for notational convenience

- What is $p(\sigma^2 \mid \mu, d_1, \dots, d_n)$?

$$L(\sigma^2) = \prod_{i=1}^n p(y_i \mid \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{d_i^2}{2\sigma^2}}$$

$$L(\sigma^2) \propto (\sigma^2)^{-n/2} \exp \left\{ -\sum_{i=1}^n d_i^2 / (2\sigma^2) \right\}, \sigma^2 > 0$$

$$p(\sigma^2) \propto \frac{1}{\sigma^2}$$

Jeffreys:

$$P(\sigma^2) \propto 1/\sigma^2$$

$$P(\sigma) \propto 1/\sigma$$

$$\underline{P(\log \sigma) \propto \text{const.}}$$

$$\int_0^{\infty} \frac{1}{x} dx = \infty$$

(improper!)

Known mean, unknown variance

- Assume we have n mean-zero normal observations with variance σ^2

Mean: $\frac{a}{b-1}$

- Jeffreys prior $p(\sigma^2) \propto \frac{1}{\sigma^2}$ (careful!)

- The posterior:

$$p(\sigma^2 \mid y) \propto (\sigma^2)^{-n/2-1} \exp \left\{ -\sum_{i=1}^n d_i^2 / (2\sigma^2) \right\}$$

- This distribution called an inverse-Gamma distribution.

- If $X \sim \text{Gamma}(a, b)$ then $\frac{1}{X} \sim \text{Inv-Gamma}(a, b)$.

$z \sim \text{Inv-Gamma}$ $z^{-1} = e^{-\frac{1}{z}}$

$\sigma^2 / y_1, \dots, y_n \sim \text{Inv-Gamma}(n/2, \frac{\sum d_i^2}{2})$

$$E[\sigma^2 | y_1, \dots, y_n, \mu] = \frac{\frac{\sum d_i^2}{2}}{n/2 - 1} = \frac{\sum d_i^2}{n-1}$$

$$= \frac{\sum (y_i - \mu)^2}{n-1}$$

(sample variance!)

A conjugate prior distribution

- In general, the conjugate prior distribution has the form:

$$p(\sigma^2) \propto (\sigma^2)^{-k_1} e^{\frac{k_2}{\sigma^2}} \quad (\text{IG Conjugate prior})$$

- Inverse-gamma: $p(y \mid a, b) = \frac{b^a}{\Gamma(a)} y^{-a-1} \exp\left(-\frac{b}{y}\right)$
 - The mean of an inverse-Gamma is $b/(a-1)$

$\Rightarrow$ IG. Posterior.

$$\text{Scaled-Inverse-}\chi^2\left(n, \frac{\sum d_i^2}{n}\right) \quad (\text{BDA3})$$



$$\text{I. G.}\left(n/2, \frac{\sum d_i^2}{2}\right)$$

$$\chi_K^2 = \sum_{i=1}^K z_i^2, \quad z_i \sim N(0,1) \quad (\text{Chi-Square})$$

$$\sigma^2 \chi_K^2 \sim \text{scaled-}\chi\text{-sq}$$

$$\chi_K^2 \equiv \text{Gamma}\left(\frac{K}{2}, 1/2\right)$$

$$(\chi_K^2)^{-1} \equiv \text{Inv. - Gamma.}$$

Joint inference for the mean and variance

In the normal model we typically factorize the prior distribution $p(\mu, \sigma^2) = p(\mu | \sigma^2) p(\sigma^2)$.

Specifically:

Normal

Inv-Gamma

$$E[\sigma^2] = \frac{\nu_0}{\nu_0 - 1} \sigma_0^2$$

$$\sigma^2 \sim \text{Inv-Gamma}(\nu_0/2, (\nu_0/2)\sigma_0^2)$$

$$\mu | \sigma^2 \sim \text{normal}(\mu_0, \sigma^2 / \kappa_0)$$

$$Y_1, \dots, Y_n | \mu, \sigma \sim \text{i.i.d. normal}(\mu, \sigma^2)$$

"Normal-inverse Gamma"

- ν_0 is interpreted as the prior sample size for σ^2

- σ_0^2 is a prior sample variance for σ^2

prior sample size for μ .
prior mean for μ

Joint inference for the mean and variance

- We write down $p(\mu, \sigma \mid y_1, \dots, y_n) \propto \underline{L(\mu, \sigma)} \underline{p(\mu, \sigma)}$. Now what?
- Estimands of potential interest:

- Monte Carlo*
- $E[\mu \mid y_1, \dots, y_n]$ *Posterior Mean of μ .*
 - $E[\sigma \mid y_1, \dots, y_n]$ *Posterior Mean of σ (std)*
 - $E[\sigma/\mu \mid y_1, \dots, y_n]$ (coefficient of variation)

Posterior mean of $\frac{\sigma}{\mu}$.

$$\int_0^{\infty} \int_{-\infty}^{\infty} \frac{\sigma}{\mu} P(\mu, \sigma \mid y_1, \dots, y_n) d\mu d\sigma$$

Posterior inference for arbitrary functions

Method of transformations

1. Find the inverse, ~~$\theta = g^{-1}(\gamma) = \frac{e^\gamma}{1+e^\gamma}$~~
2. Compute $\frac{dg^{-1}(\gamma)}{d\gamma}$

$$\gamma = \frac{\sigma}{\mu} = g(\mu, \sigma)$$

3. Find

$$\underline{p_\gamma(\gamma \mid y_1, \dots, y_n)} = \left| \frac{dg^{-1}(\gamma)}{d\gamma} \right| \times p_\theta(g^{-1}(\gamma) \mid y_1, \dots, y_n)$$

$$'' \int \delta P_\gamma(\gamma \mid y_1, \dots, y_n) d\gamma.$$

Posterior inference for arbitrary functions

For any $\gamma = g(\theta)$ we have

$$E(\underbrace{g(\theta)}_{\gamma} | y) = \int \underbrace{g(\theta)}_{\gamma} p(\theta | y) d\theta = \int \gamma \overbrace{p(\gamma | y_1, \dots, y_n)}^{\text{HARD!}} d\gamma.$$

Examples:

- $E[\gamma | y] = \int \log\left(\frac{\theta}{1-\theta}\right) p(\theta | y) d\theta$
- $Pr(\theta \in R | y) = E(I[\theta \in R] | y)$

$$\int \underbrace{g(\theta)}_{\gamma} p_{\theta}(\underbrace{\theta}_{\text{Easy}} | y_1, \dots, y_n) d\theta = \int \gamma \underbrace{p_{\gamma}(\gamma | y_1, \dots, y_n)}_{\text{hard}} d\gamma$$

Law of the Unconscious Statistician



Monte Carlo Method

iid draws from $P(\theta | y_1, \dots, y_n)$

- $\bar{\theta} = \sum_{s=1}^S \theta^{(s)} / S \rightarrow E[\theta | y_1, \dots, y_n]$
LLN
- $\sum_{s=1}^S (\theta^{(s)} - \bar{\theta})^2 / (S - 1) \rightarrow \text{Var}[\theta | y_1, \dots, y_n]$
- $\# (\theta^{(s)} \leq c) / S \rightarrow \Pr(\theta \leq c | y_1, \dots, y_n)$
Number of
- the α -percentile of $\{\theta^{(1)}, \dots, \theta^{(S)}\} \rightarrow \theta_\alpha$ *(quantiles)*

Almost sure convergence due to the SLLN

Strong Law
of Large Numbers

Get S samples from

$P(\theta | y_1, \dots, y_n)$, use to approximate $E[g(\theta) | y_1, \dots, y_n]$

Monte Carlo Error

S iid samples, then

$$\frac{\sum_{i=1}^S g(\Theta_i)}{S} \xrightarrow{\text{distr.}} \mathcal{N}\left(E[g(\vartheta) | y_1, \dots, y_n], \frac{\text{Var}(g(\vartheta) | y_1, \dots, y_n)}{S}\right)$$

(Central Limit Theorem)

Back to the normal model

Example: midge wing length

- Modeling wing length of different species of midge (small, two-winged flies)
- From prior studies: mean wing length close to 1.9mm. μ_0
- Prior mean for μ is $\mu_0 = 1.9$ and Jeffreys prior for σ^2 .
- Prior sample sizes: choose $\kappa_0 = 1$ $P(\sigma^2) \propto \frac{1}{\sigma^2}$
- $(\bar{y}, s^2) = (1.804, 0.0169)$ are the sufficient statistics

$$P(\mu, \sigma^2) = P(\mu | \sigma^2) P(\sigma^2)$$
$$\downarrow \qquad \qquad \qquad \downarrow$$
$$N(1.9\text{mm}, \frac{\sigma^2}{1}) \quad 1/\sigma^2$$

Working with the log posterior

- As always, we will write down $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$
- In code, we always work with the log-posterior for numerical reasons
 - Mathematically it makes no difference, but computationally it is important
 - $L(\theta) \propto \prod p(y_i \mid \theta)$ is very small for moderate sample size (underflow)
 - $\ell(\theta) = \sum \log(p(y_i \mid \theta))$ is numerically stable
- Monte Carlo methods only require that we can evaluate the log posterior

Grid approximation

```
1 log_normal_posterior <- Vectorize(function(mu, sigma) {  
2   (unnormalized)  
3   ### log likelihood  
4   sum(dnorm(y, mu, sigma, log=TRUE)) +  
5   ## plus log prior  
6   dnorm(mu, mu0, sigma/sqrt(k0), log=TRUE) +  
7   -2*log(sigma)  
8 }  
9 )
```

μ σ

$\ell(\mu, \sigma) +$

$\log P(\mu, \sigma)$

$P(\sigma^2) \propto \frac{1}{\sigma^2}$

```
10  
11  
12 post_grid <- as_tibble(  
13   expand_grid(seq(1.6, 2.0, by=0.001),  
14     seq(0.01, 0.25, by=0.001))  
15   colnames(post_grid) <- c("mu", "s")
```

$\rightarrow [\text{vec 1}, \text{vec 2}]$

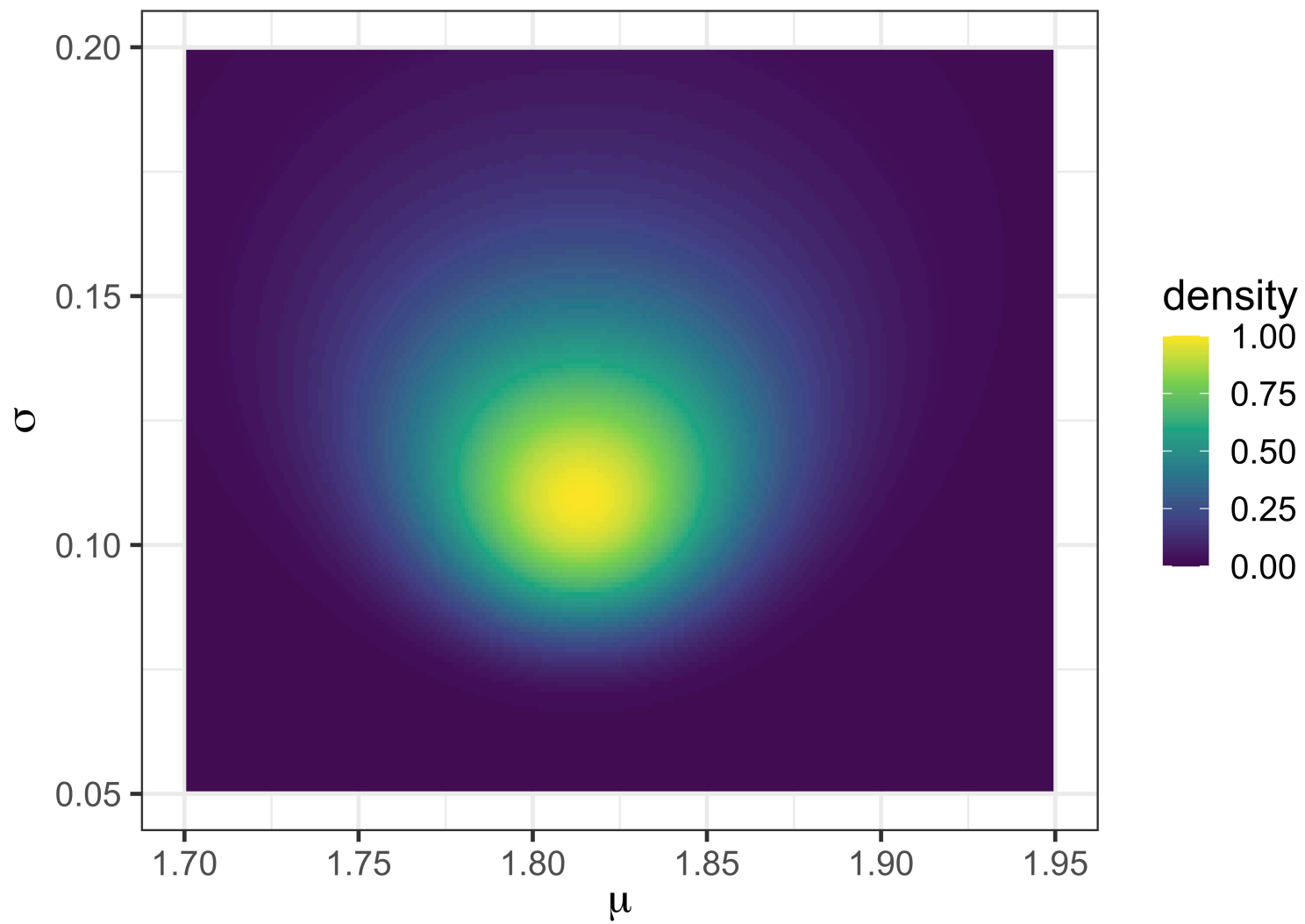
μ σ

1.6	.01
1.6	.02
1.6	.03
...	...
1.8	.25
1.8	.25

Grid approximation

```
1 post_grid %>%  
2   mutate(log_density = log_normal_posterior(mu, s)) %>%  
3   mutate(density = exp(log_density - max(log_density))) %>%  
4   ggplot() +  
5   geom_raster(aes(mu, s, fill=density)) +  
6   xlim(c(1.7, 1.95)) + ylim(c(0.05, 0.2)) +  
7   xlab(expression(mu)) +  
8   ylab(expression(sigma)) +  
9   theme_bw(base_size=16  
10             ) +  
11   scale_fill_continuous(type="viridis")
```

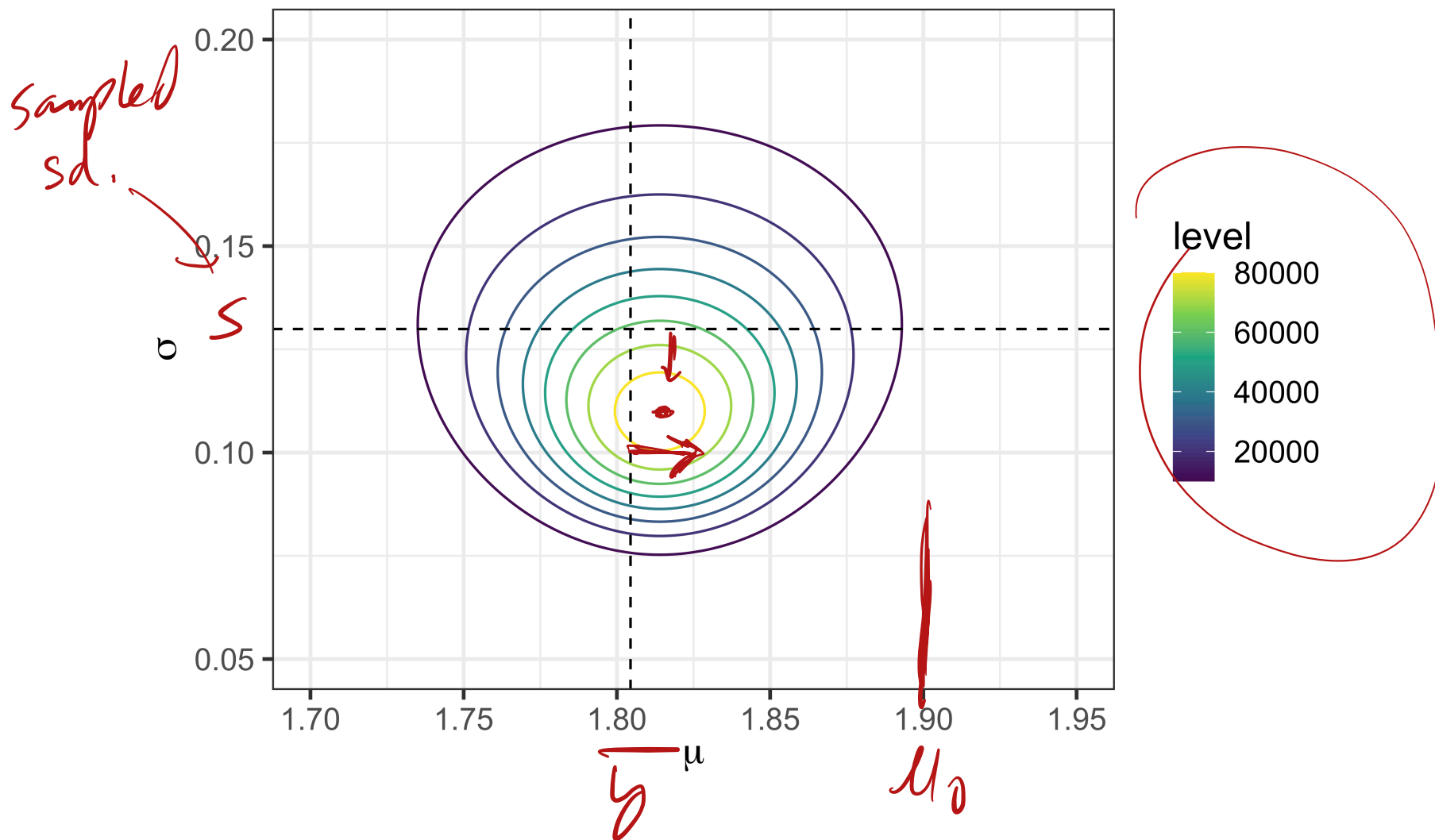
Unnormalized!



Contour Plot (Standard Deviation)

```
1 post_grid %>%
2   mutate(density = exp(log_normal_posterior(mu, s))) %>%
3   ggplot() +
4   geom_contour(aes(mu, s, z=density, colour=stat(level))) +
5   xlim(c(1.7, 1.95)) + ylim(c(0.05, 0.2)) +
6   xlab(expression(mu)) + ylab(expression(sigma)) +
7   ggtitle("Posterior Contours") +
8   theme_bw(base_size=16) +
9   scale_color_continuous(type="viridis") +
10  geom_hline(yintercept=s, linetype="dashed") +
11  geom_vline(xintercept=ybar, linetype="dashed")
```

Posterior Contours

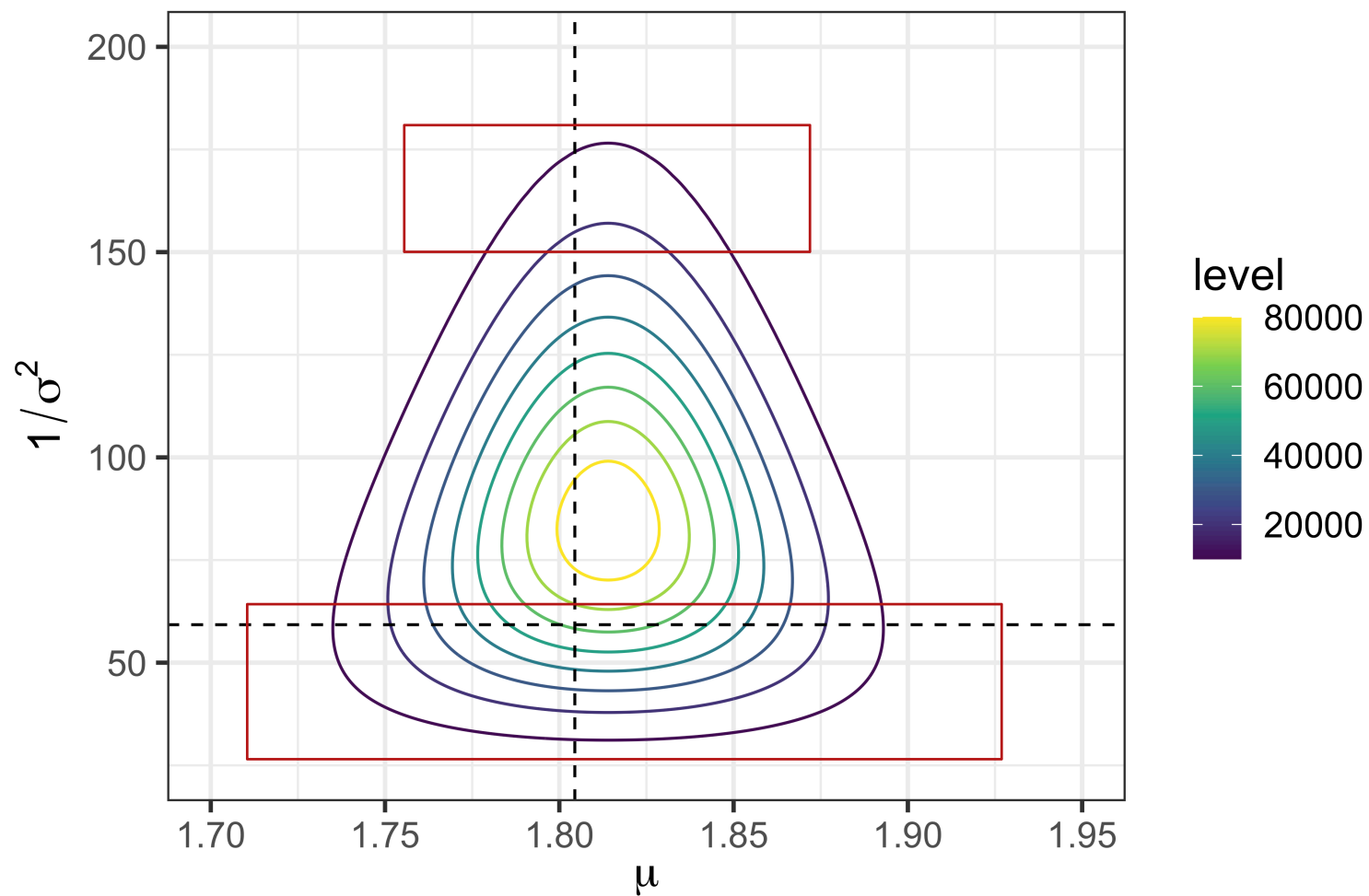


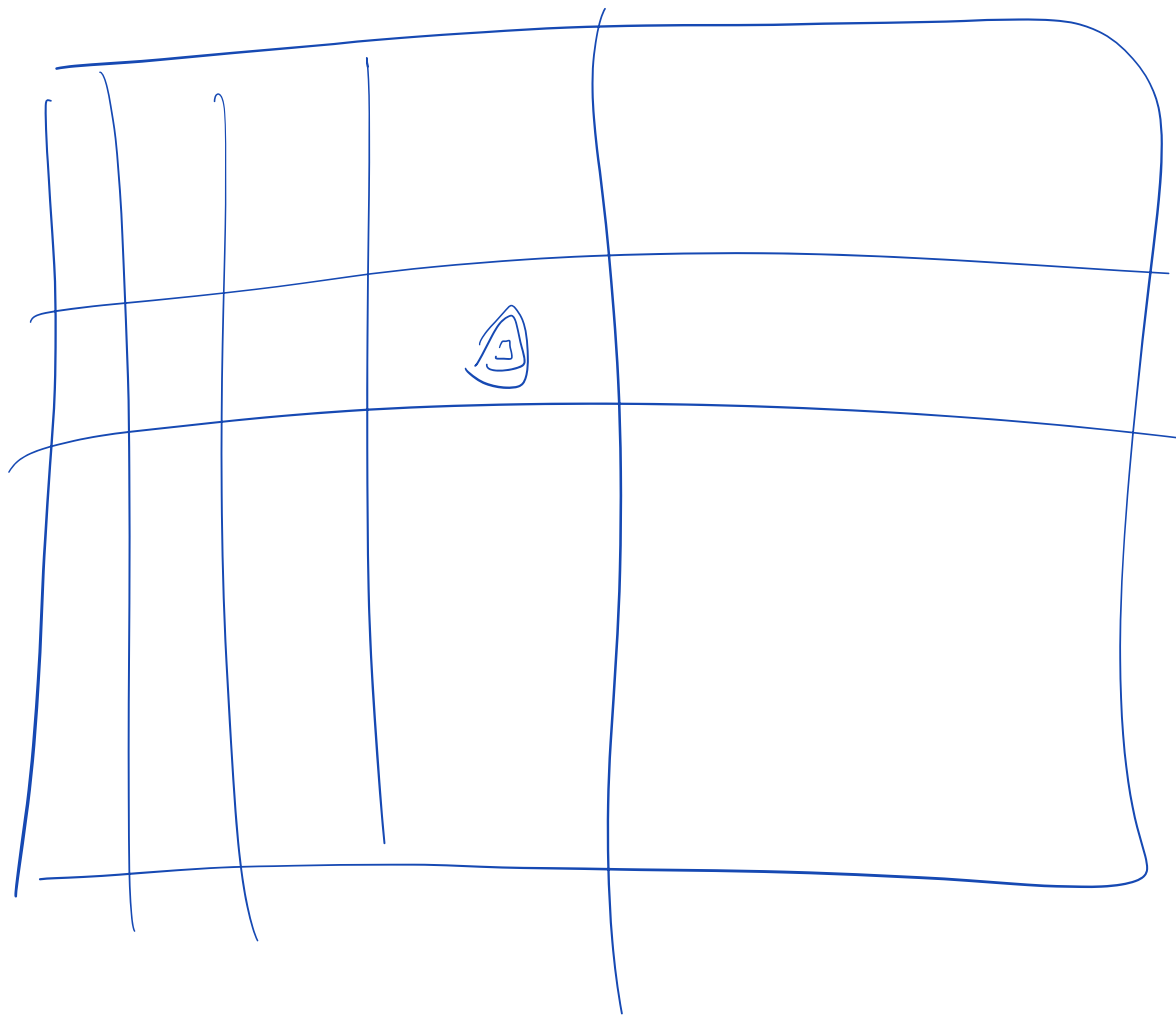
Contour Plot (Precision)

$$\sigma \rightarrow \frac{1}{\sigma^2}$$

```
1 post_grid$prec <- 1/post_grid$s^2
2
3 post_grid %>%
4   mutate(density = exp(log_normal_posterior(mu, sqrt(1/prec)))) %>%
5   ggplot() +
6     geom_contour(aes(mu, prec, z=density, colour=stat(level))) +
7     xlim(c(1.7, 1.95)) + ylim(c(25, 200)) +
8     xlab(expression(mu)) + ylab(expression(1/sigma^2)) +
9     ggtitle("Posterior Contours") +
10    theme_bw(base_size=16) +
11    scale_color_continuous(type="viridis") +
12    geom_hline(yintercept=1/s^2, linetype="dashed") +
13    geom_vline(xintercept=ybar, linetype="dashed")
```

Posterior Contours





Sampling from the joint posterior

- Contour and raster plots allow us to visualize the posterior (in two dimensions)
 - Grid sampling approximates posterior samples
 - Need to know approximately where the high posterior density is (not easy)
- When we have more than 2 parameters visualization isn't feasible
- How do we summarize the posterior?
 - e.g. posterior means, posterior probabilities, intervals, etc..

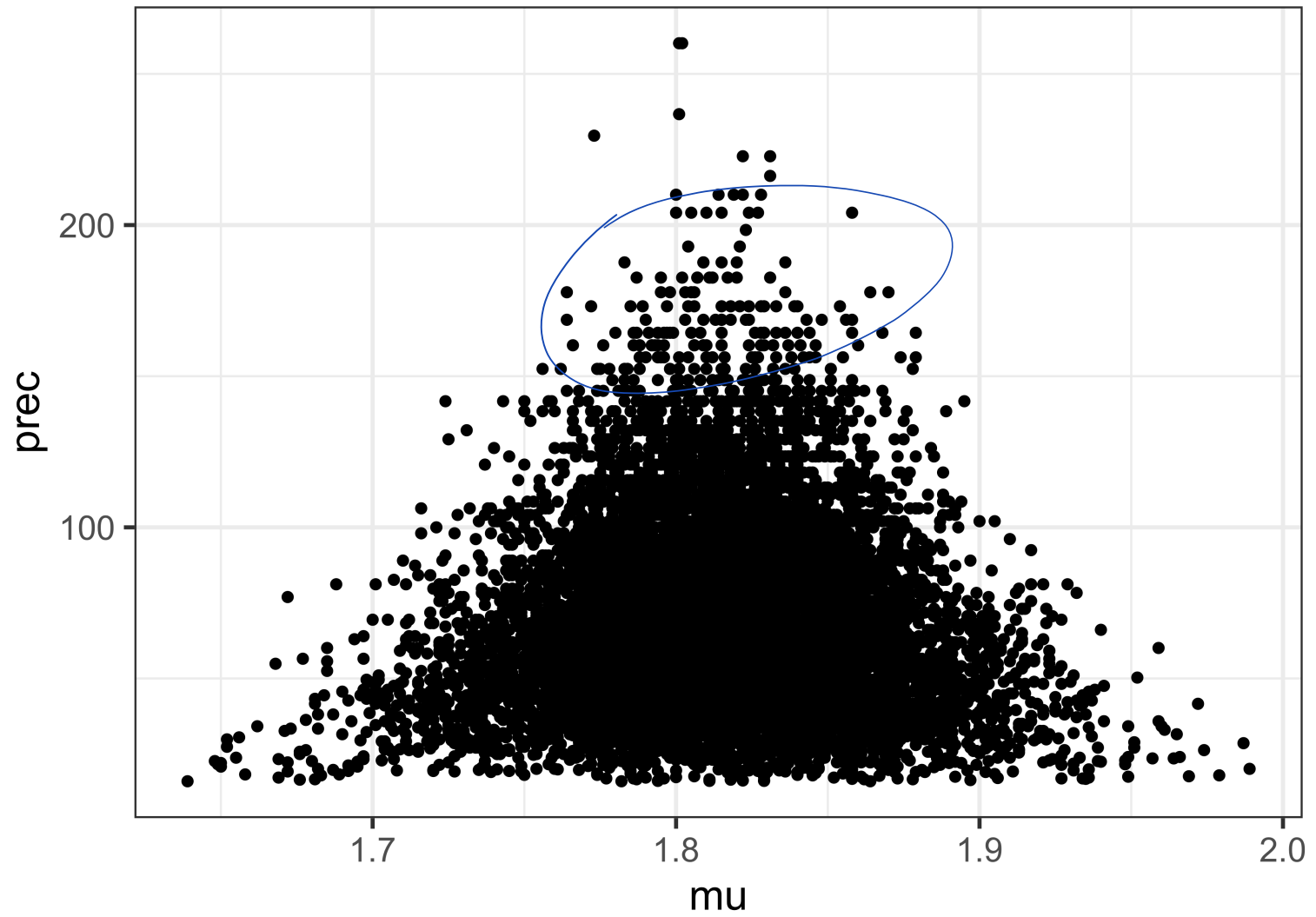
Visualizing Posterior Samples

```
1 post_grid %>%  
2   mutate(density = exp(log_normal_posterior(mu, sqrt(1/prec)))) %>%  
3   mutate(density = density / sum(density)) -> post_grid (normalization)  
4  
5 sample_indices <- sample(1:nrow(post_grid), size=10000, replace=TRUE, prob=  
6  
7 post_grid[sample_indices, ] %>%  
8   mutate(prec = 1/s^2) %>%  
9   ggplot() +  
10  geom_point(aes(x=mu, y=prec)) +  
11  theme_bw(base_size=16)
```

Density

Grid
Sampling

	.01	.01		
		.02		
		.05	.07	

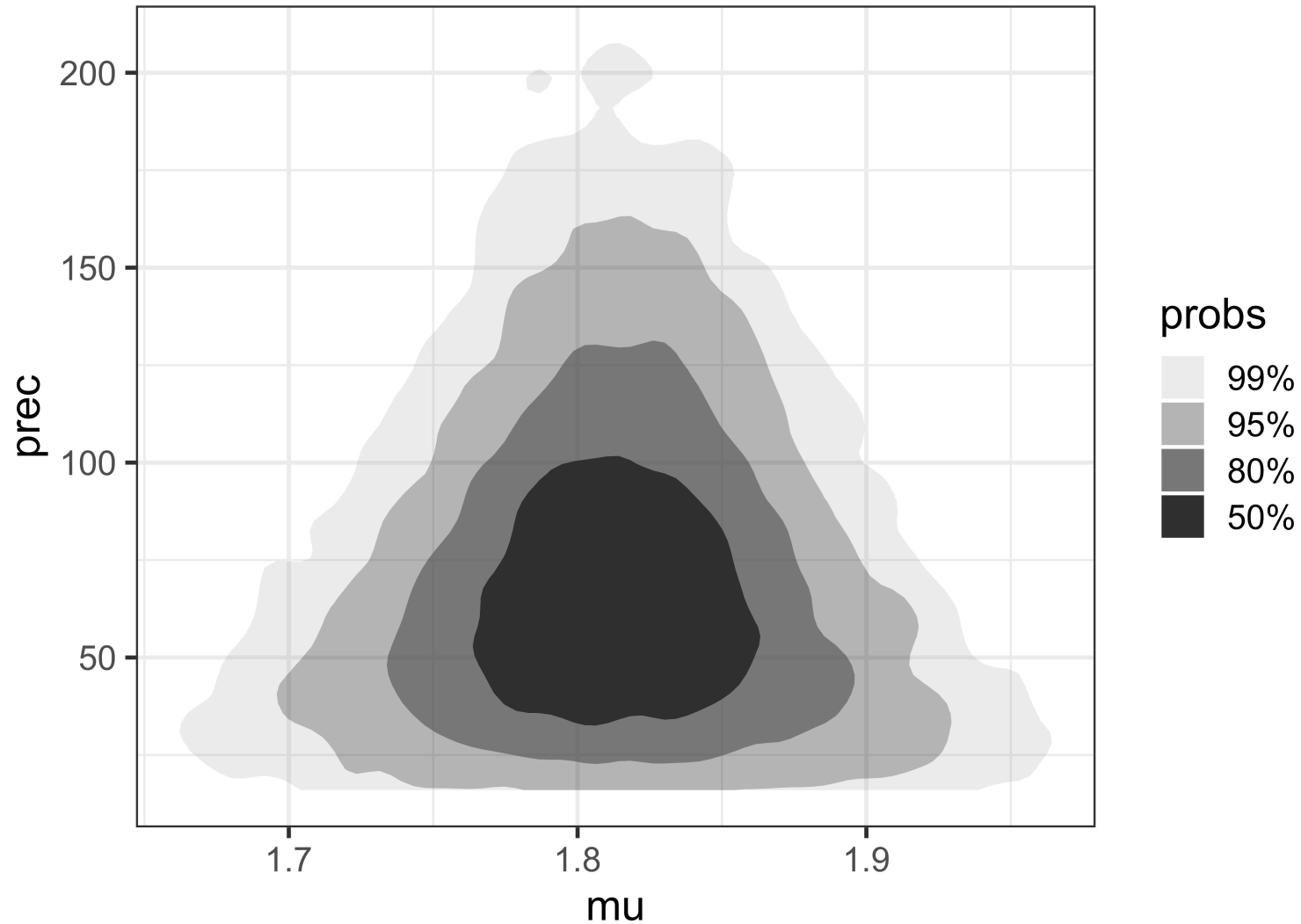


Visualizing Posterior Samples

```
1 post_grid %>%
2   mutate(density = exp(log_normal_posterior(mu, sqrt(1/prec)))) %>%
3   mutate(density = density / sum(density)) -> post_grid
4
5 sample_indices <- sample(1:nrow(post_grid), size=10000, replace=TRUE, prob=
6
7 post_grid[sample_indices, ] %>%
8   mutate(prec = 1/s^2) %>%
9   ggplot() +
10  ggdensity::geom_hdr(aes(x=mu, y=prec)) +
11  theme_bw(base_size=16)
```

Highest Density Region!

Visualizing Posterior Samples



Direct Sampling

Normal (rnorm(-, ...))

- $p(\mu, \sigma^2 | y) = p(\mu | \sigma^2, y) p(\sigma^2 | y)$
- Need to know how to sample from $p(\sigma^2 | y) = \int p(\mu, \sigma^2 | y) d\mu$ and $p(\mu | \sigma^2, y)$ "Marginal Posterior of σ^2 "
- With the proposed conjugate priors, this integral is tractable

$$P(\sigma^2 | y_1, \dots, y_n) \sim \text{Inv-Gamma}(n/2, \frac{\nu_n \sigma_n^2}{2})$$

$$\nu_n = \nu_0 + n$$

$$\sigma_n^2 = \frac{1}{\nu_n} \left[\nu_0 \sigma_0^2 + (n-1) s^2 + \frac{k_0 n}{n+k_0} (\bar{y} - \mu_0)^2 \right]$$

$x \leftarrow \text{gamma}(-, \cdot), \text{ return } \frac{1}{x}$

Direct Sampling

- $p(\mu, \sigma^2 \mid y) = p(\mu \mid \sigma^2, y)p(\sigma^2 \mid y)$
- Need to know how to sample from $p(\sigma^2 \mid y) = \int p(\mu, \sigma^2 \mid y)d\mu$ and $p(\mu \mid \sigma^2, y)$
- With the proposed conjugate priors, this integral is tractable

$\{1/\sigma^2 \mid y_1, \dots, y_n\} \sim \text{gamma}(\nu_n/2, \nu_n\sigma_n^2/2)$, where

$$\nu_n = \nu_0 + n$$

$$\sigma_n^2 = \frac{1}{\nu_n} \left[\nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n} (\bar{y} - \mu_0)^2 \right]$$

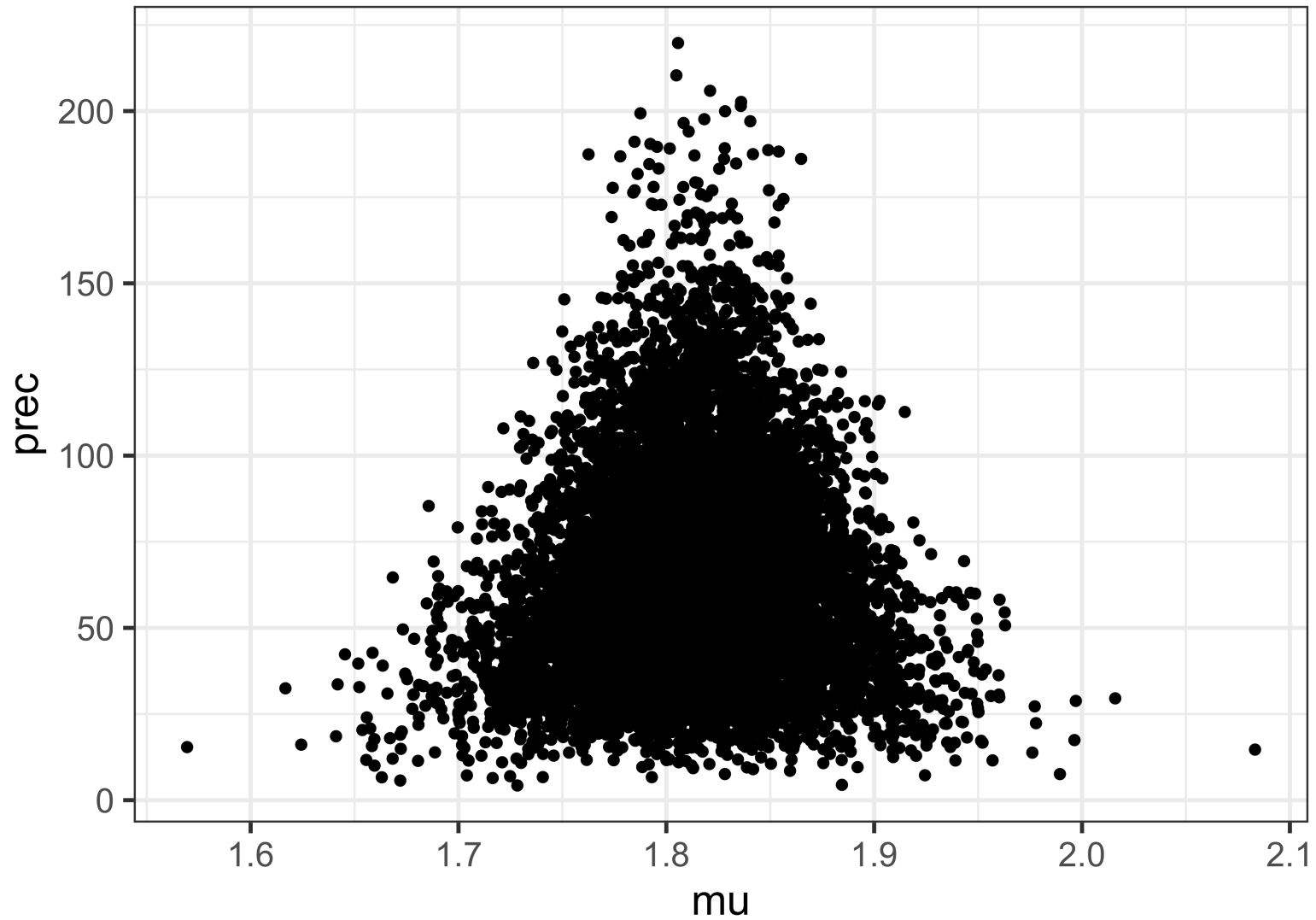
Direct Sampling

```

1 ## posterior parameters
2 kn <- k0 + n
3 nun <- nu0 + n
4 mun <- (k0 * mu0 + n * ybar) / kn
5 s2n <- (nu0*s20 + (n-1)*s2 + k0*n / kn * (ybar - mu0)^2) / nun
6
7 nsamps <- 10000
8 prec_samps <- rgamma(nsamps, nun/2, nun*s2n/2)
9 mu_samps <- rnorm(nsamps, mun, sqrt((1/prec_samps) / kn))
10
11 tibble(mu=mu_samps, prec=prec_samps) %>%
12   ggplot() +
13   geom_point(aes(x=mu, y=prec)) +
14   #ggdensity::geom_hdr(aes(x=mu, y=prec)) +
15   theme_bw(base_size=16)

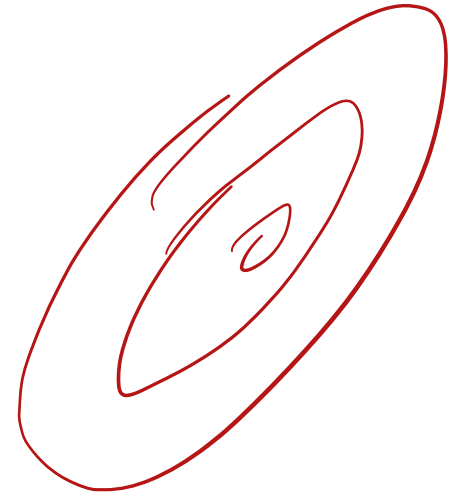
```

Direct Sampling



The Multivariate Normal Distribution

$$\underline{Y}_{p \times 1} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_p \end{bmatrix} \sim N_p(\underline{\mu}, \underline{\Sigma})$$



$$P(\underline{\mu}, \underline{\Sigma} \mid \underline{Y})$$

$$P(\underline{Y} = \underline{y} \mid \underline{\mu}, \underline{\Sigma}) = (2\pi)^{-p/2} |\underline{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2} (\underline{y} - \underline{\mu})' \underline{\Sigma}^{-1} (\underline{y} - \underline{\mu})\right]$$

Matrix quadratic

$$\underline{\mu} \in \mathbb{R}^p$$

$$\underline{\Sigma} \in S_p^+ \text{ "cone of positive definite matrices"}$$

$$a' \Sigma a > 0 \quad \forall a \in \mathbb{R}^p$$

Σ has $\frac{p(p+1)}{2}$ "free" parameters

$$P(\mu \mid \Sigma, y_1, \dots, y_n)$$

$$p(\mu) \sim N_p(\underline{\mu}_0, \underline{\Lambda}_0)$$

Prior covariance for μ .

Conjugate prior for μ .

$$L(\mu) \propto \prod_{i=1}^n (2\pi)^{-1/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2}(y_i - \mu)' \Sigma^{-1}(y_i - \mu)\right]$$

$$\propto \exp\left(-\frac{1}{2} \sum_{i=1}^n (y_i - \mu)' \Sigma^{-1}(y_i - \mu)\right)$$

(check)

$$\propto \exp\left(-\frac{1}{2} \mu' A \mu + \mu' b\right)$$

$$A = n \Sigma^{-1} \quad b = n \Sigma^{-1} \bar{y}$$

$$P(\mu) \propto \exp(\mu' \Lambda_0^{-1} \mu + \mu' \Lambda_0^{-1} \mu_0)$$

$$P(\mu | \dots) \propto L(\mu) P(\mu)$$

$$\propto \exp(\mu' A_n \mu + \mu' b_n)$$

$$A_n = \Sigma^{-1} + \Lambda_0^{-1}$$

$$b_n = n \Sigma^{-1} \bar{y} + \Lambda_0^{-1} \mu_0$$

$$P(\mu | y_1, \dots, y_n) \sim N(\overset{\text{post mean}}{\mu_n}, \overset{\text{post cov. of } \mu}{\Sigma_n})$$

$$\Sigma_n = (n \Sigma^{-1} + \Lambda_0^{-1})^{-1}$$

$$\mu_n = (n \Sigma^{-1} + \Lambda_0^{-1})^{-1} (n \Sigma^{-1} \bar{y} + \Lambda_0^{-1} \mu_0)$$

Special case

Assume I choose $\Lambda_0^{-1} = \kappa_0 \Sigma^{-1}$

Then:

$$\mu_n = (n \Sigma^{-1} + \kappa_0 \Sigma^{-1})^{-1} (n \Sigma^{-1} \bar{y} + \kappa_0 \Sigma^{-1} \mu_0)$$

$$= \frac{1}{n + \kappa_0} \Sigma (n \Sigma^{-1} \bar{y} + \kappa_0 \Sigma^{-1} \mu_0)$$

$$= \left(\frac{n}{n + \kappa_0} \bar{y} + \frac{\kappa_0}{n + \kappa_0} \mu_0 \right) \in \mathcal{R}^p$$

Prior for Σ ?

Inv-Wishart Prior.

(Generalizes Gamma/Inv-Gam
to multiple dimensions)

Dirichlet-Multinomial

(Generalizes Beta-Binomial)

Metagenomics example

- Metagenomics is the study of genetic material recovered directly from environmental samples
- Map counts of genetic material to counts of microbial species
- Assume species are sampled with replacement
 - Observed sample is a multinomial distribution
- Total counts isn't meaningful (hard to control how much total sample)
- Relative counts are meaningful

Multinomial Density

- Let $\theta = (\theta_1, \theta_2, \dots, \theta_k)$ with $\sum_i \theta_i = 1$
- If $Y = (Y_1, \dots, Y_K) \sim \text{Mult}(n, \theta)$, then:
 - $Y_i \sim \text{Bin}(n, \theta_i)$
 - $Y_i + Y_j \sim \text{Bin}(n, \theta_i + \theta_j)$
- What is $\hat{\theta}_{MLE}$?

Dirichlet Distribution

Bayes Estimators

Why the posterior mean?

- Often times we need to make a “decision” by providing a single estimate
- The posterior provides a full distribution over θ , which can be summarized in infinitely many ways
- Specify a *loss function* which describes the cost of estimating $\hat{\theta}$ when the truth is θ

Bayes Estimators

- The *loss function*: $L(\hat{\theta}, \theta)$
 - Squared error: $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$
 - Absolute error: $L(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$
- The *Bayes risk* is the posterior expected loss: $E_{\theta|y}[L(\hat{\theta}, \theta)]$
- Summarize the posterior by minimizing the Bayes risk.
- An estimator $\hat{\theta}$ is said to be a Bayes estimator if it minimizes the Bayes risk among all estimators.

Examples Squared error loss

$$\min_{\hat{\theta}} E_{\theta|y}(\hat{\theta} - \theta)^2 = \min_{\hat{\theta}} \int (\hat{\theta} - \theta)^2 p(\theta | y) d\theta$$

The Bias-Variance Tradeoff

- The prior distribution (usually) makes your estimator biased...
- But the prior distribution also (usually) reduces the variance!
- Bayes estimators are (usually) consistent
- Example: compute the frequentist mean and variance of the posterior mean.

Example: IQ scores

- Scoring on **IQ tests** is designed to yield a $N(100, 15)$ distribution for the general population
- We observe IQ scores for a sample of n individuals from a particular town and estimate μ , the town-specific IQ score
- If we lacked knowledge about the town, a natural choice would be $\mu_0 = 100$
- Suppose the true parameters for this town are $\mu = 112$ and $\sigma = 13$
 - The town is smarter on average than the general population

Example: IQ scores

- What is the mean squared error of the MLE? MSE of the posterior mean?

$$\bullet \text{MSE} [\hat{\mu}_{MLE}] = \text{Var} [\hat{\mu}_{MLE}] = \frac{\sigma^2}{n} = \frac{169}{n}$$

$$\bullet \text{MSE} [\hat{\mu}_{PM} | \theta_0] = w^2 \frac{169}{n} + (1 - w)^2 144$$

Example: IQ scores

